

Question 1

Design the circuit in figure below to obtain a dc voltage of 0 V at each of the drains of Q_1 and Q_2 when $v_{G1}=v_{G2}=0$ V. Operate all transistors at $V_{ov}=0.15$ V and assume that for the process technology in which the circuit is fabricated, $V_{tn}=0.35$ V and $\mu_n C_{ox}=400 \mu\text{A}/\text{V}^2$. Neglect channel-length modulation. Determine the values of R , R_D , and the W/L ratios of Q_1 , Q_2 , Q_3 and Q_4 . What is the input common-mode voltage range for your design?

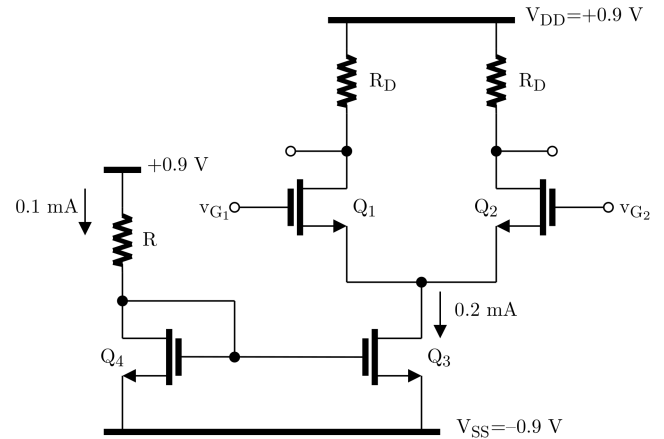
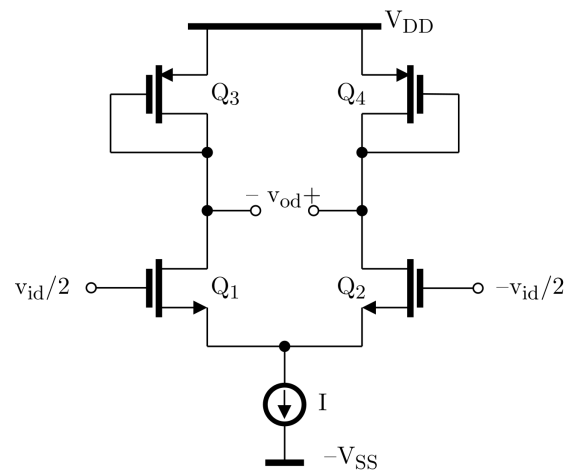
**Question 2**

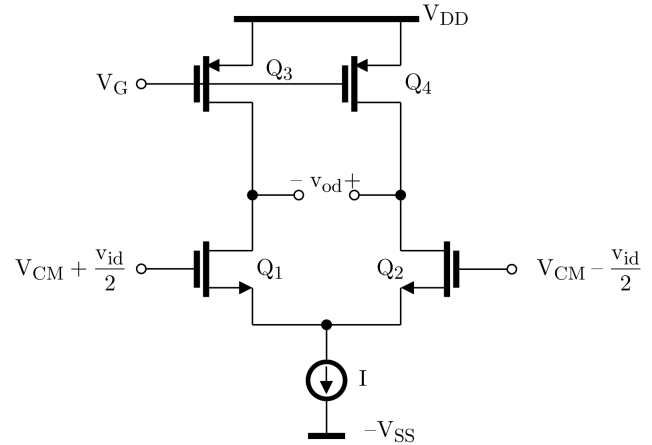
Figure below shows a MOS differential amplifier with the drain resistors R_D implemented using diode-connected PMOS transistors, Q_3 and Q_4 . Let Q_1 and Q_2 be matched, and Q_3 and Q_4 be matched.

- Find the differential half-circuit and use it to derive an expression for A_d in terms of $g_{m1,2}$, $g_{m3,4}$, $r_{o1,2}$, and $r_{o3,4}$.
- Neglecting the effect of the output resistances r_o , find A_d in terms of μ_n , μ_p , $(W/L)_{1,2}$ and $(W/L)_{3,4}$.
- If $\mu_n=4\mu_p$ and all four transistors have the same channel length, find $(W_{1,2}/W_{3,4})$ that results in $A_d = 10 \text{ V/V}$.



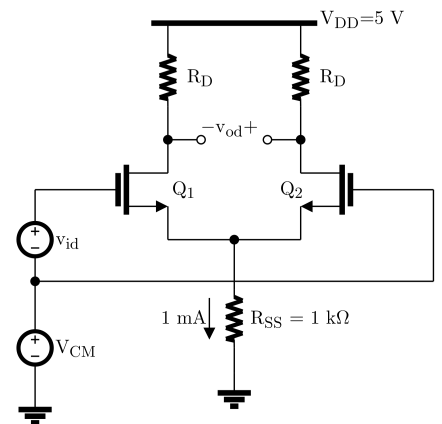
Question 3

The differential amplifier of figure below is fabricated in a $0.18\text{-}\mu\text{m}$ CMOS technology for which $\mu_n C_{ox} = 4\mu_p C_{ox} = 400 \mu\text{A}/\text{V}^2$, $|V_t| = 0.5 \text{ V}$, and $|V'_A| = 10 \text{ V}/\mu\text{m}$. If the bias current $I = 200 \mu\text{A}$ and all transistors have a channel length twice the minimum and are operating at $|V_{OV}| = 0.2 \text{ V}$, find W/L for each of Q_1 , Q_2 , Q_3 , and Q_4 , and determine the differential voltage gain A_d .

**Question 4**

The differential amplifier in figure below utilizes a resistor R_{SS} to establish a 1-mA DC bias current. Note that this amplifier uses a single 5-V supply and thus the DC common-mode voltage V_{CM} cannot be zero. Transistors Q_1 and Q_2 have $k'_n W/L = 2.5 \text{ mA}/\text{V}^2$, $V_t = 0.7 \text{ V}$, and $\lambda = 0$.

- Find the required value of V_{CM} .
- Find the value of R_D that results in a differential gain A_d of $8 \text{ V}/\text{V}$.
- Determine the DC voltage at the drains.
- Determine the single-ended-output common-mode gain $\Delta V_{D1}/\Delta V_{CM}$. (Hint: You need to take $1/g_m$ into account.)
- Use the common-mode gain found in (d) to determine the change in V_{CM} that results in Q_1 and Q_2 entering the triode region.



Question 5

In a current-mirror-loaded differential amplifier of the form shown in figure below, all transistors are characterized by $k'W/L=4 \text{ mA/V}^2$, and $|V_A|=5 \text{ V}$. Find the bias current I for which the gain $v_o/v_{id}=20 \text{ V/V}$.

